10. You are given a long conducting wire, a pair of slab-shaped magnets on steel yoke and a light-beam galvanometer for detecting small currents. With the aid of a diagram, describe an experiment to investigate **TWO** factors affecting the e.m.f. induced in a conductor when it moves in a magnetic field. (7 marks)

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*4. An electron moving with speed 1.2×10^7 m s^{-1} enters a square region $ABCD$ with a uniform magnetic field of 0.01 T pointing into the paper as shown in Figure 4.1. The electron describes a quarter circle from A to C and it emerges from C with the same speed. Neglect the effects of gravity.

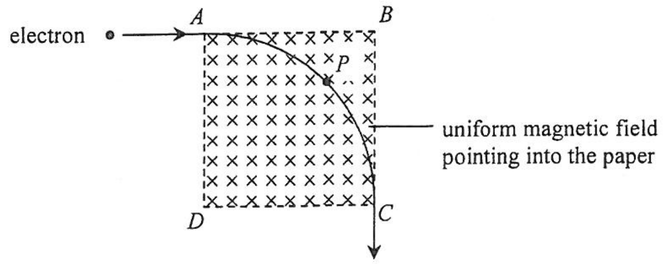

Figure 4.1

(a) (i) Find the magnitude of the magnetic force acting on the electron at point P on its path. (2 marks)

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(ii) Indicate in Figure 4.1 the direction of the electron's acceleration at the point P. (1 mark)

(b) Although the electron accelerates due to the magnetic force, explain why it emerges from the magnetic field with the same speed. (2 marks)

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(c) Deduce the speed of the electron when entering the magnetic field such that it would describe a semi-circle from A to D instead. (2 marks)

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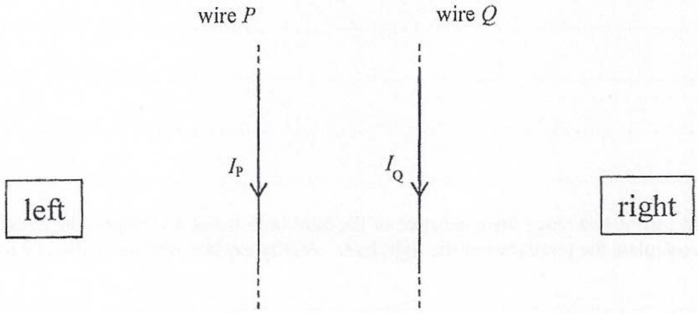
9. (a) Two long straight current carrying wires, P and Q, are parallel to each other and lie on the plane of the paper as shown in Figure 9.1. The currents in the wires, I_P and I_Q, flow in the same direction.

Figure 9.1



(i) State the direction (to the left / to the right / into the paper / out of the paper) of the magnetic field at Q due to P. (1 mark)

....

....

(ii) In Figure 9.1, draw the direction of the magnetic force acting on Q due to P. (1 mark)

(iii) Show that the magnitude of the magnetic force per unit length F_l acting on Q due to P is

$$F_l = \frac{\mu_o I_P I_Q}{2\pi r},$$

where μ_o is the permeability of free space and r is the separation between the two wires. (3 marks)

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(iv) For the magnetic force acting on Q due to P and the magnetic force acting on P due to Q, if $I_P \neq I_Q$, briefly explain whether the two forces are equal in magnitude. (2 marks)

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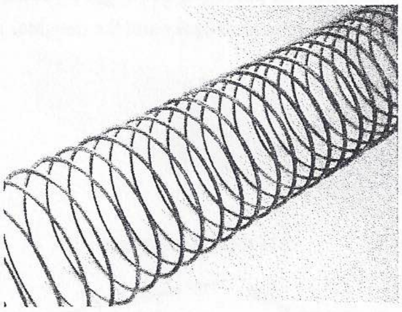
(b) Figure 9.2 shows a metal slinky spring.

Figure 9.2



(i) If a direct current passes through the spring, briefly explain whether the spring will be compressed or stretched due to magnetic force. (2 marks)

(ii) A student suggests that the spring will be compressed and stretched alternately due to magnetic force when an alternating current passes through. Briefly explain why he is wrong. (1 mark)

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9. A rectangular coil *PQRS* of 20 turns, each having an area of 0.005 m^2, is placed in a uniform magnetic field *B* of strength 0.3 T pointing into the paper as shown in Figure 9.1.

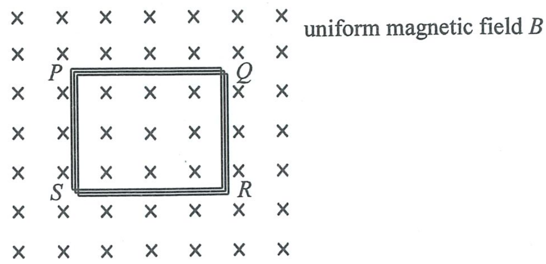

Figure 9.1

(a) The strength of the magnetic field decreases uniformly to zero within 0.5 s.

(i) Explain why a current would be induced in the coil. (2 marks)

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*(ii) Calculate the change in total magnetic flux linkage through the coil and the value of the induced e.m.f. ξ in the coil. (3 marks)

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(b) Now the coil is rotated uniformly about an axis through 180° as shown in Figures 9.2(a) and 9.2(b) within 0.5 s.

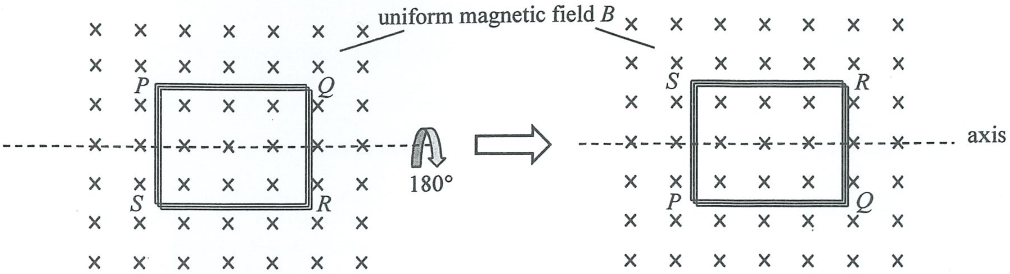

Figure 9.2(a)

Figure 9.2(b)

(i) State the value of the change in total magnetic flux linkage through the coil in this case (1 mark)

....

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(ii) At the moment when the coil rotated through 90°, would the induced current flow in the direction *PQRS*, *PSRQ* or is there no induced current in the coil ? (1 mark)

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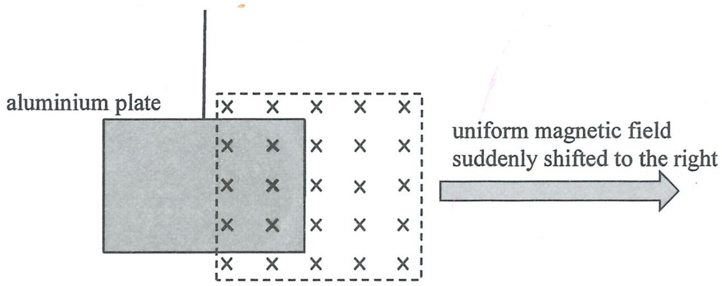
(c) Figure 9.3 shows a thin rectangular aluminium plate suspended by a long string. The plate is partly inside a uniform magnetic field provided by a strong magnet.

Figure 9.3



The magnet, which is not in contact with the plate, is suddenly shifted to the right.

(i) On Figure 9.3, draw a small circle at the location where eddy currents are induced on the aluminium plate. Use an arrow to indicate the direction of current. (2 marks)

(ii) Describe the subsequent motion of the aluminium plate, if any. (1 mark)

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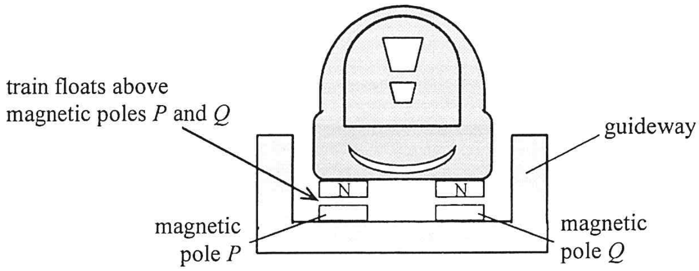
3. Read the following passage about a **magnetically levitated (maglev) train** and answer the questions that follow.

'A maglev train car is just a box with magnets on the four corners,' says Jesse Powell, the son of the maglev train inventor. The electromagnets employed have superconducting coils (i.e. coils with extremely low resistance). They therefore can generate magnetic fields 10 times stronger than ordinary electromagnets, enough to levitate and propel a train.



LEVITATION

Two sets of magnetic fields are set up for different functions. One is to make the train float a few centimetres above magnetic poles *P* and *Q* as shown while the other is a propulsion system run by an alternating current for moving the train car along the guideway by magnetic attraction and repulsion. This floating design enables a smooth movement of the train. Even when the train travels up to 600 km per hour, passengers inside experience less vibration than travelling on traditional trains.

(a) Explain why electromagnets employing superconducting coils can produce much stronger magnetic fields. (2 marks)

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Please stick the barcode label here.

(b) State the polarities of the magnetic poles P and Q and explain how this arrangement enables the train to float. (2 marks)

(c) Referring to the resistive forces experienced by the train, explain why a maglev train ride is (i) smoother and (ii) faster. (2 marks)

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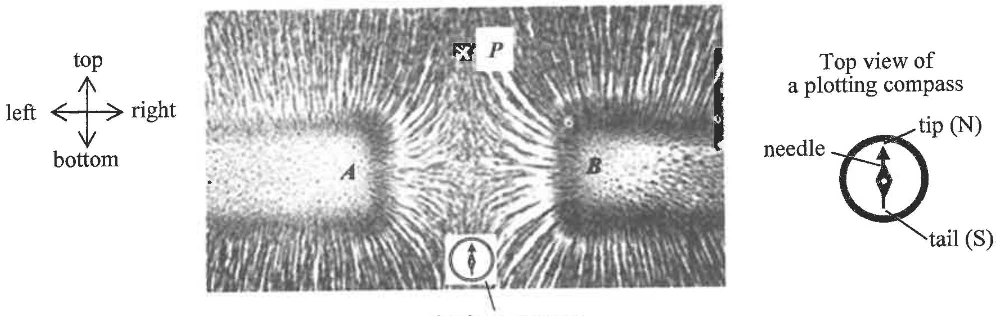
7. Read the following passage about **magnetic field patterns** and answer the questions that follow.

Iron filings are tiny pieces of iron nearly in powder form. Since iron is ferromagnetic, a magnetic field would induce each piece of iron filings to become a 'tiny bar magnet'. The south pole of each of these 'tiny bar magnets' attracts the north poles of neighbouring iron filings. The magnetic field pattern is displayed as iron filings align themselves with the field lines.

The figure below shows such a pattern displayed on a cardboard with two identical bar magnets placed underneath. A plotting compass placed at the bottom of the figure points to the top as shown.



(a) (i) State the polarity of the respective poles at A and B of the bar magnets. (1 mark)

A : ....

B : ....

(ii) If the compass is moved to P, towards what direction (top, bottom, left or right) would it be pointing ? (1 mark)

....

....

(iii) In obtaining such field patterns experimentally, one is advised to place the magnets **underneath the cardboard**. Why ? (1 mark)

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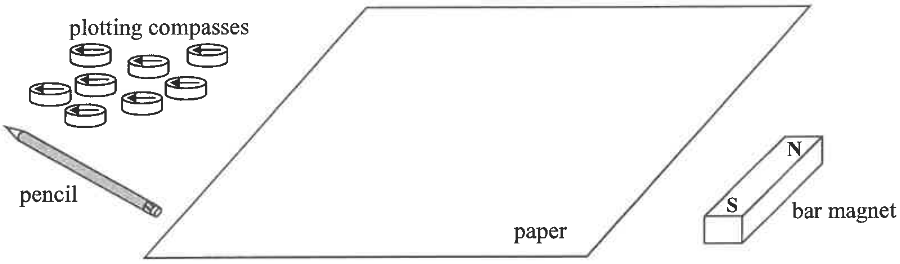
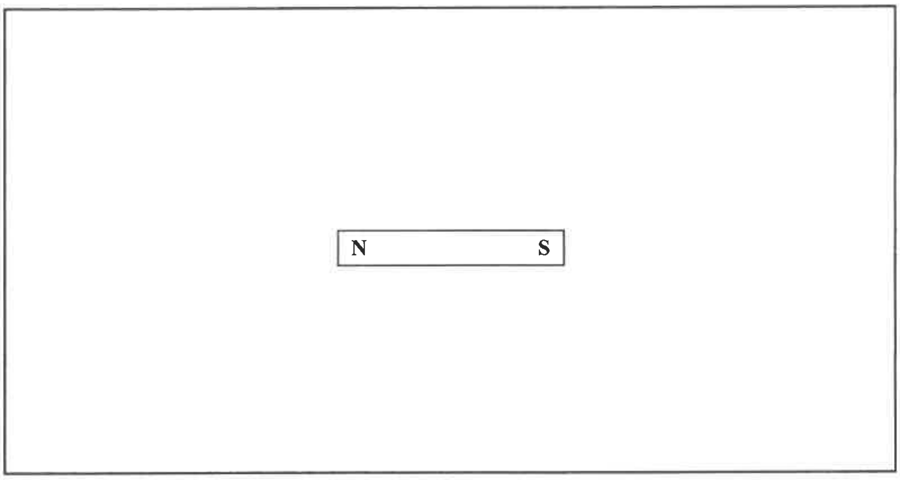
(b) You are given a bar magnet (with polarities marked), 8 small plotting compasses, a pencil and a piece of white paper as shown in Figure 7.1

Figure 7.1



(i) Describe, with the aid of a diagram, how you would use the apparatus given to trace several field lines around the bar magnet. Neglect the Earth's magnetic field. (5 marks)

Top view



(ii) Suggest **ONE** advantage of using a plotting-compass method over an iron-filing method in studying magnetic fields. (1 mark)

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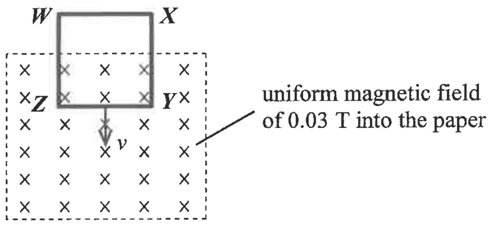
9. A square metal loop $WXYZ$ of side length 0.10 m is inserted with a constant velocity v into a uniform magnetic field of flux density 0.03 T. The magnetic field is perpendicular to the plane of the loop as shown in Figure 9.1. The resistance of **each side** of the metal loop is 0.15 Ω.

Figure 9.1



When the metal loop is entering the magnetic field, a current of 0.01 A flows in the loop.

(a) On Figure 9.1, indicate the direction of this current. (1 mark)

*(b) Find v. (2 marks)

(c) (i) Find the **potential difference** V_{YZ} between Y and Z. (2 marks)

(ii) Explain whether V_{YZ} is equal to the **induced e.m.f.** across YZ. (1 mark)

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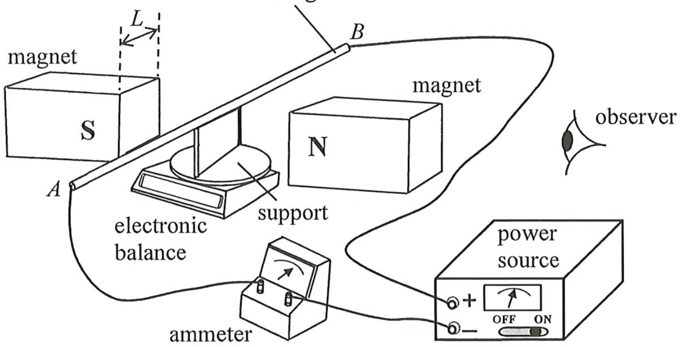
8. A student conducts an experiment to study the force on a straight current-carrying wire placed between the poles of a magnet providing a uniform magnetic field perpendicular to the wire. An adjustable power source is connected to the wire which is held horizontally by an insulated support placed on an electronic balance. The variation of the magnetic force on the wire is registered by the balance.

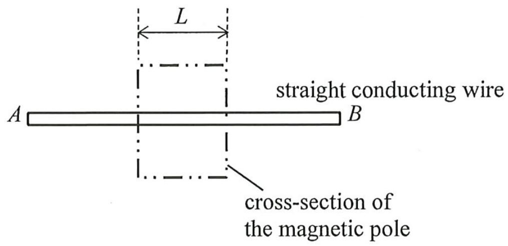
Figure 8.1



(a) On Figure 8.1, mark the positive (+) and negative (–) terminals of the ammeter. (1 mark)

(b) Figure 8.2 shows the side view of the wire AB seen by the observer. On the figure, draw the magnetic field provided by the magnet and indicate the directions of the current I and magnetic force F_B on the wire. (3 marks)

Figure 8.2

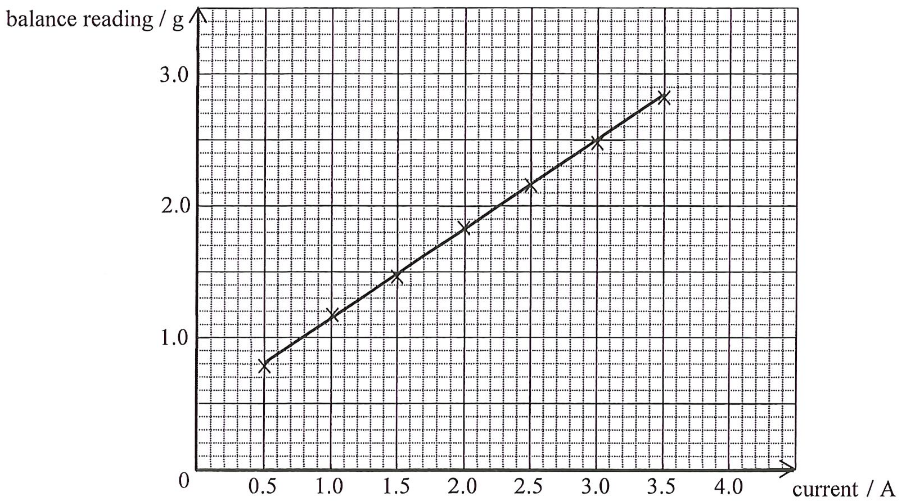


(c) Complete the table below to indicate how each of the factors listed would affect the magnitude of the magnetic force F_B (increase, decrease or unchanged). (3 marks)

	magnetic force F_B
using a stronger magnet of the same dimensions	
using another support such that the position of the horizontal wire is slightly lowered	
using a longer wire while the current is kept unchanged	

Answers written in the margins will not be marked.

(d) The graph below is plotted by using the experimental data obtained.



(i) Describe the corresponding experimental procedures to be carried out. (2 marks)

(ii) Describe how the balance reading varies with the current as revealed by the graph. (1 mark)

(iii) Explain why the graph does not pass through the origin when the current is zero. (1 mark)

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*8. A rectangular coil, with N turns and each turn of area 9.00×10^{-4} m^2, is placed in a uniform magnetic field of strength B. Initially the plane containing the coil is parallel to the magnetic field. The coil is then rotated steadily about the axis XY as shown in Figure 8.1(a).

Figure 8.1(a)

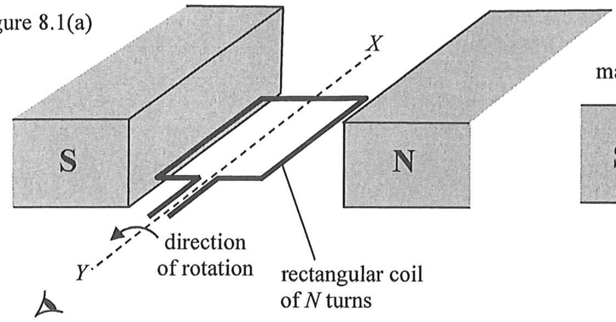
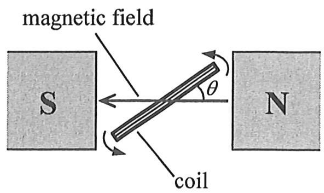
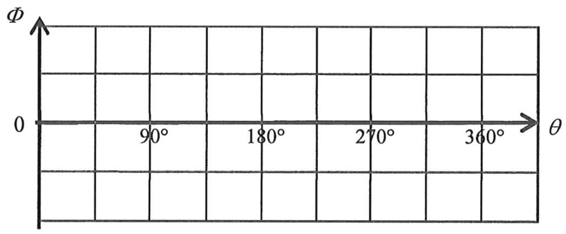


Figure 8.1(b)



(a) Sketch the variation of magnetic flux Φ passing through each turn of the coil when the coil rotates from $\theta = 0°$ to $360°$ (see Figure 8.1(b)). (2 marks)



(b) The maximum magnetic flux through each turn of the coil is 2.70×10^{-4} Wb.

(i) Determine B (unit: T). (2 marks)

(ii) Calculate the average induced e.m.f. ε in the coil when it is rotated from $\theta = 0°$ to $90°$ in 0.01 s. Given that the number of turns N is 100. (2 marks)

Answers written in the margins will not be marked.